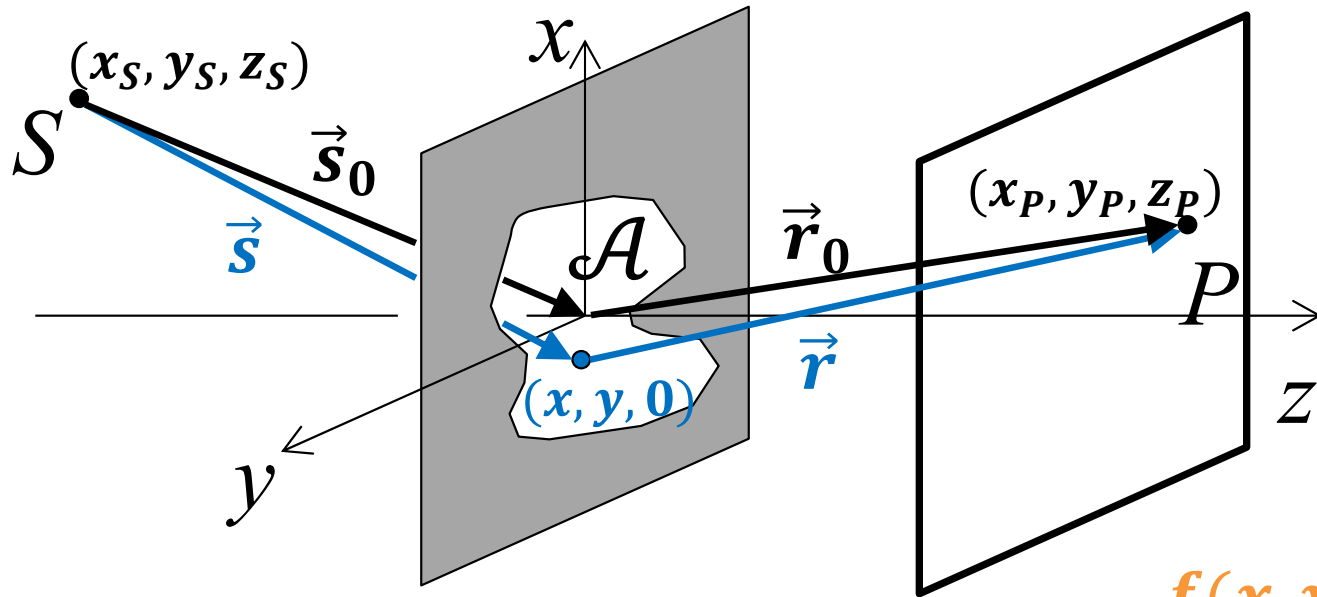


# **Ph134, Lecture 17**

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- **Fraunhofer and Fresnel diffraction: physical meanings of the terms**
- **Fraunhofer diffraction:**
  - **conditions for it**
  - **relation to Fourier transforms**
  - **1-D slit and circular aperture**
- **(Resolution and resolving power of optical instruments )**

# Fraunhofer vs. Fresnel Diffraction:



$$u_{\mathcal{A}}(P) \approx \frac{-iA_0^{(+)} e^{iks_0}}{\lambda s_0} \frac{e^{ikr_0}}{r_0} \frac{[\hat{s}_0 \cdot \hat{z} + \hat{r}_0 \cdot \hat{z}]}{2} \iint_{\mathcal{A}} dx dy e^{ik(s - s_0 + r - r_0)}$$

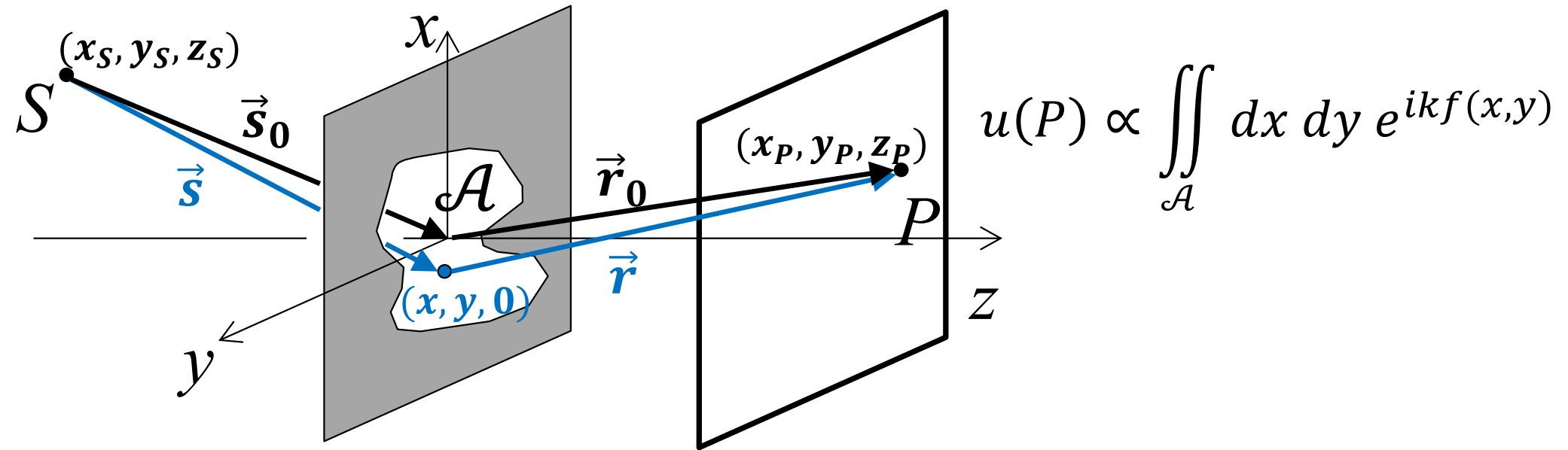
$f(x, y) \equiv (s - s_0 + r - r_0)$

$$f(x, y) \approx -\left(\frac{x_S}{s_0} + \frac{x_P}{r_0}\right)x - \left(\frac{y_S}{s_0} + \frac{y_P}{r_0}\right)y + \frac{1}{2}\left(\frac{1}{s_0} + \frac{1}{r_0}\right)(x^2 + y^2) - \frac{1}{2s_0}\left(\frac{x_S}{s_0}x + \frac{y_S}{s_0}y\right)^2 - \frac{1}{2r_0}\left(\frac{x_P}{r_0}x + \frac{y_P}{r_0}y\right)^2$$

Keep all these terms  $\rightarrow$  Fresnel diffraction.

Keep only linear terms in  $x, y \rightarrow$  Fraunhofer diffraction.

# Physical significance of the terms:



*Optical path length change due to:*

$$f(x, y) \approx -\left(\frac{x_S}{s_0} + \frac{x_P}{r_0}\right)x - \left(\frac{y_S}{s_0} + \frac{y_P}{r_0}\right)y$$

*Planar wavefronts at an angle to the aperture plane.*

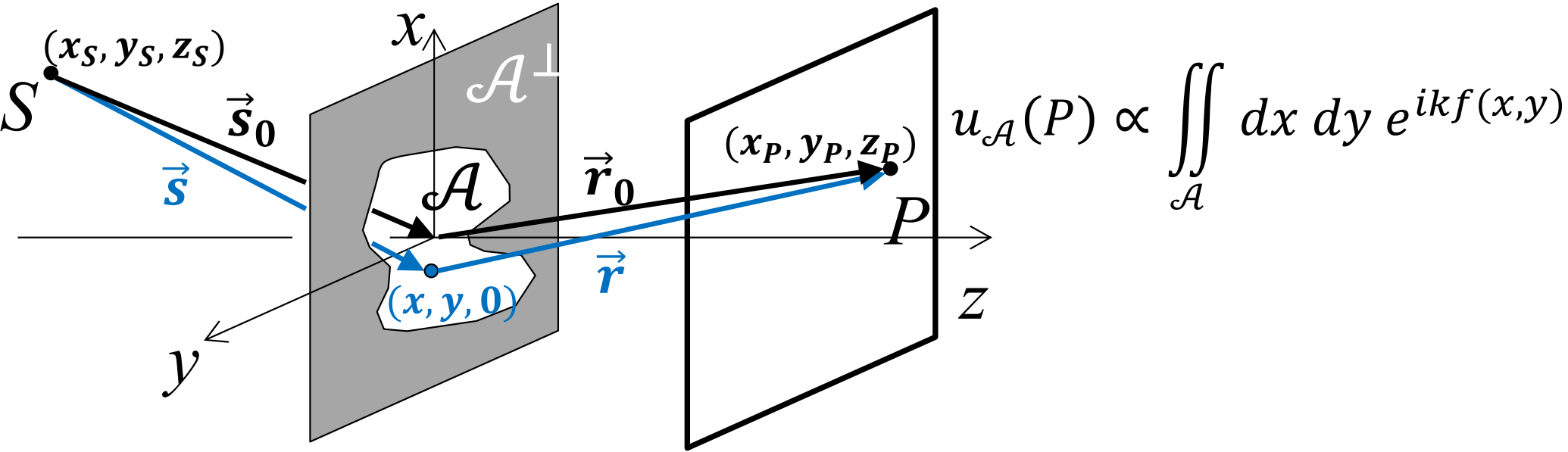
$$+ \frac{1}{2} \left( \frac{1}{s_0} + \frac{1}{r_0} \right) (x^2 + y^2)$$

*Curved wavefronts at normal incidence.*

$$- \frac{1}{2s_0} \left( \frac{x_S}{s_0}x + \frac{y_S}{s_0}y \right)^2 - \frac{1}{2r_0} \left( \frac{x_P}{r_0}x + \frac{y_P}{r_0}y \right)^2$$

*Curved wavefronts at an angle to the aperture plane.*

# Conditions for Fraunhofer diffraction:



*Neglect phase shift due to quadratic terms of  $f(x,y)$ .*

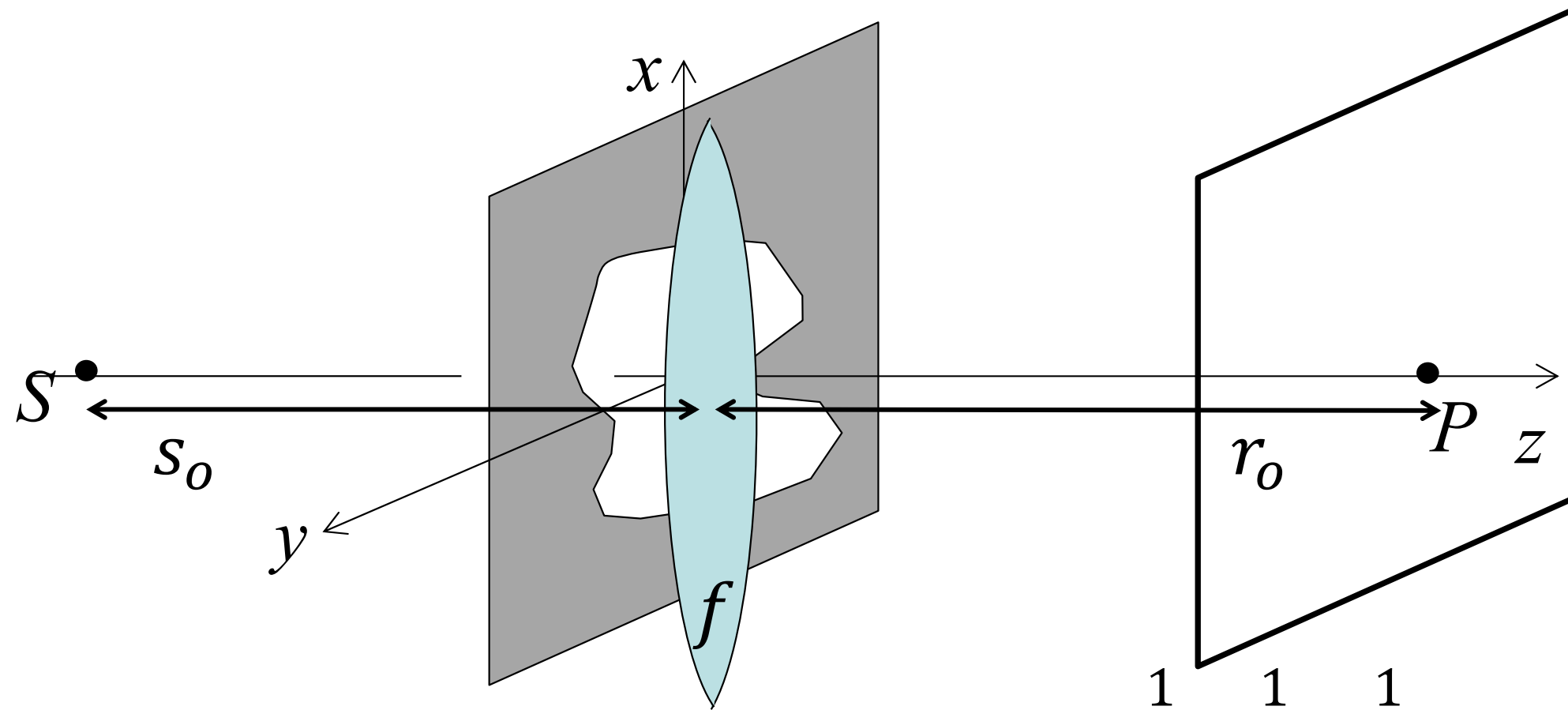
*Source location/properties:  
not a pt. source in the first  
place  
OR*

*Observation plane  
location:*

$$\frac{(\text{aperture size})^2}{\lambda r_0} \ll 1$$

$$k \frac{1}{2} \frac{(\text{aperture size})^2}{s_0} \ll \pi \iff \frac{(\text{aperture size})^2}{\lambda s_0} \ll 1$$

# Alternate condition for Fraunhofer diffraction:



*Aperture points far from  $z$  axis have optical path larger because of wavefront curvature*

*Aperture points far from  $z$  axis have optical path smaller because of small travel through glass*

$$\frac{1}{f} = \frac{1}{s_0} + \frac{1}{r_0}$$
*Effects cancel:  
wavefront curvature effects can be neglected even with small longitudinal distances!*

# Fraunhofer Diffraction and Fourier Transforms:

Define transmission function  $T(x,y)$  in aperture plane.

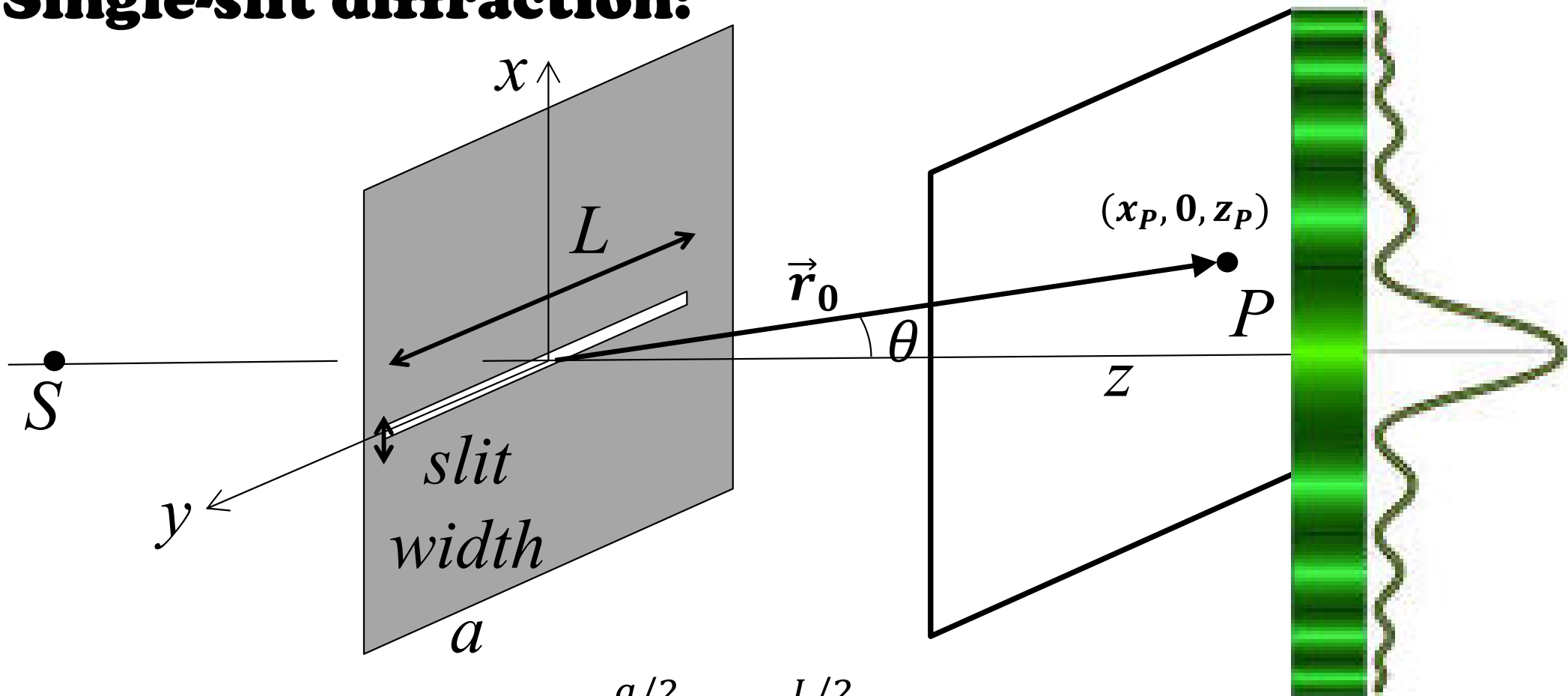
$$f(x,y) \equiv (\mathbf{s} - \mathbf{s}_0 + \mathbf{r} - \mathbf{r}_0)$$

$$u_{\mathcal{A}}(P) \approx \frac{-iA_0^{(+)} }{\lambda} \frac{e^{iks_0}}{s_0} \frac{e^{ikr_0}}{r_0} \frac{[\hat{\mathbf{s}}_0 \cdot \hat{\mathbf{z}} + \hat{\mathbf{r}}_0 \cdot \hat{\mathbf{z}}]}{2} \iint_{\mathcal{A}} dx dy e^{ik(s-s_0+r-r_0)}$$

$$f(x,y) \approx -\left(\frac{x_s}{s_0} + \frac{x_P}{r_0}\right)x - \left(\frac{y_s}{s_0} + \frac{y_P}{r_0}\right)y$$

$$u_{\mathcal{A}}(x_P,y_P) \approx (\text{constant}) \int_{-\infty}^{\infty} dx \int_{-\infty}^{\infty} dy T(x,y) e^{-ik\frac{x_P}{r_0}x} e^{-ik\frac{y_P}{r_0}y}$$

# Single-slit diffraction:



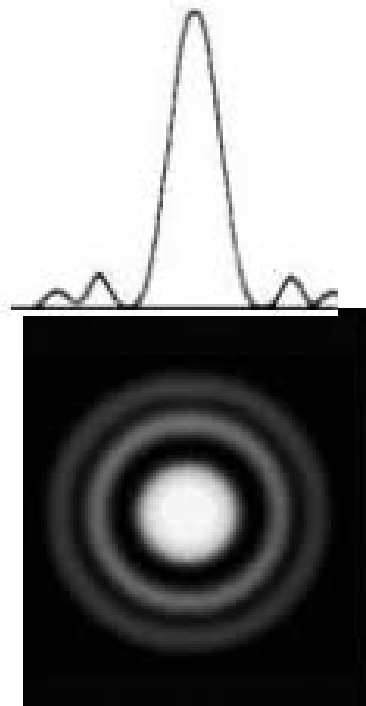
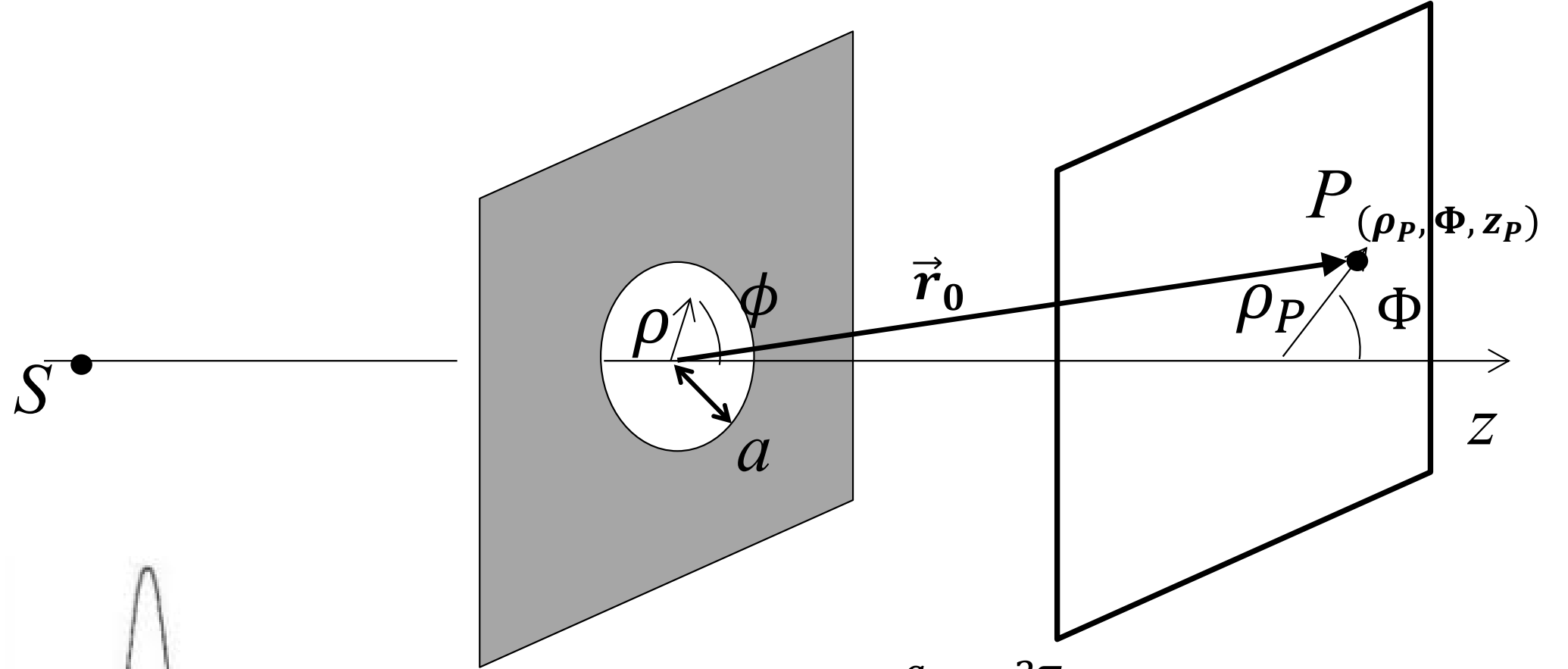
$$u_{\mathcal{A}}(x_P, 0) \approx (\text{constant}) \int_{-a/2}^{a/2} dx \int_{-L/2}^{L/2} dy e^{-ik \frac{x_P}{r_0} x}$$

$$= (\text{constant}) La \frac{\sin \alpha}{\alpha}$$

$$\alpha \equiv \frac{\pi a \sin \theta}{\lambda}$$

$$\text{first min at } \sin \theta = \frac{\lambda}{a}$$

# Circular-aperture diffraction:

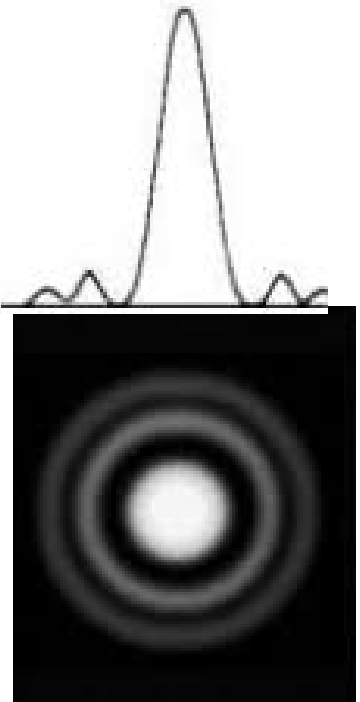
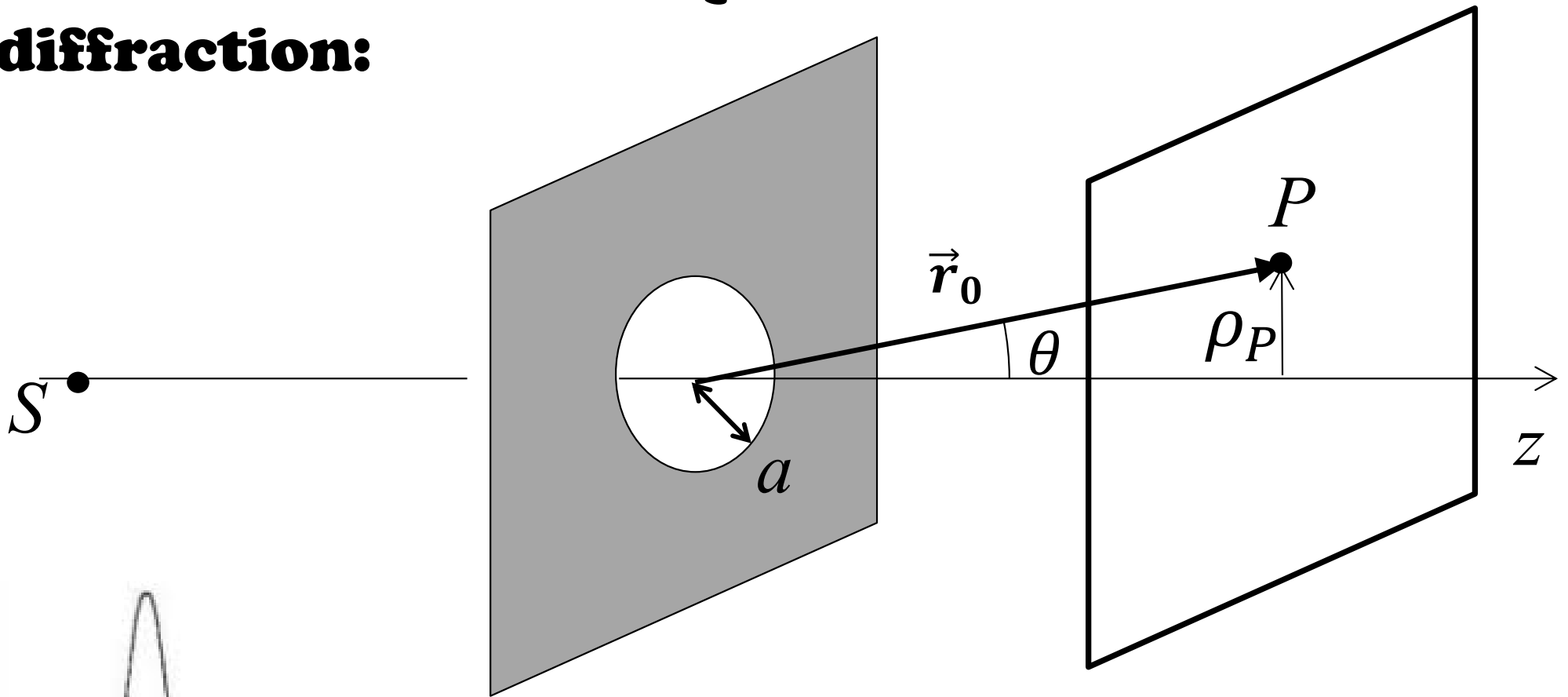


$$u_{\mathcal{A}}(\rho_P, \Phi_P) \approx (\text{constant}) \int_0^a d\rho \int_0^{2\pi} d\phi \, \rho \, e^{-ik \frac{\rho_P}{r_0} \rho \cos(\phi - \Phi)}$$

*Bessel functions!*  
*Airy diffraction pattern*

*first min at*  $\frac{\rho_P}{r_0} = \frac{1.22\lambda}{(2a)}$

# Fraunhofer circular-aperture diffraction:

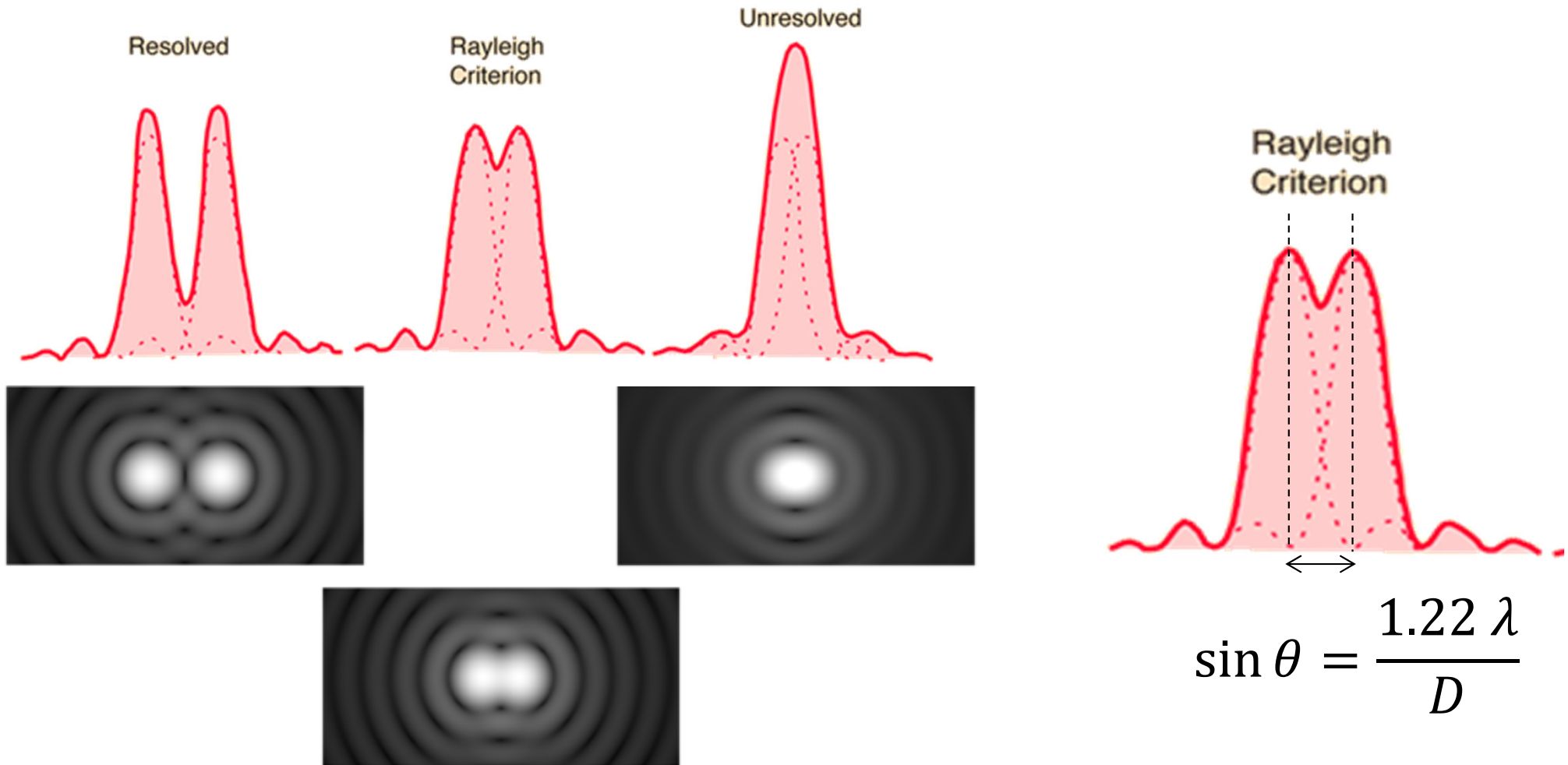


*Bessel functions!*  
*Airy diffraction pattern*

$$\text{first min at } \frac{\rho_P}{r_0} = \frac{1.22\lambda}{(2a)}$$

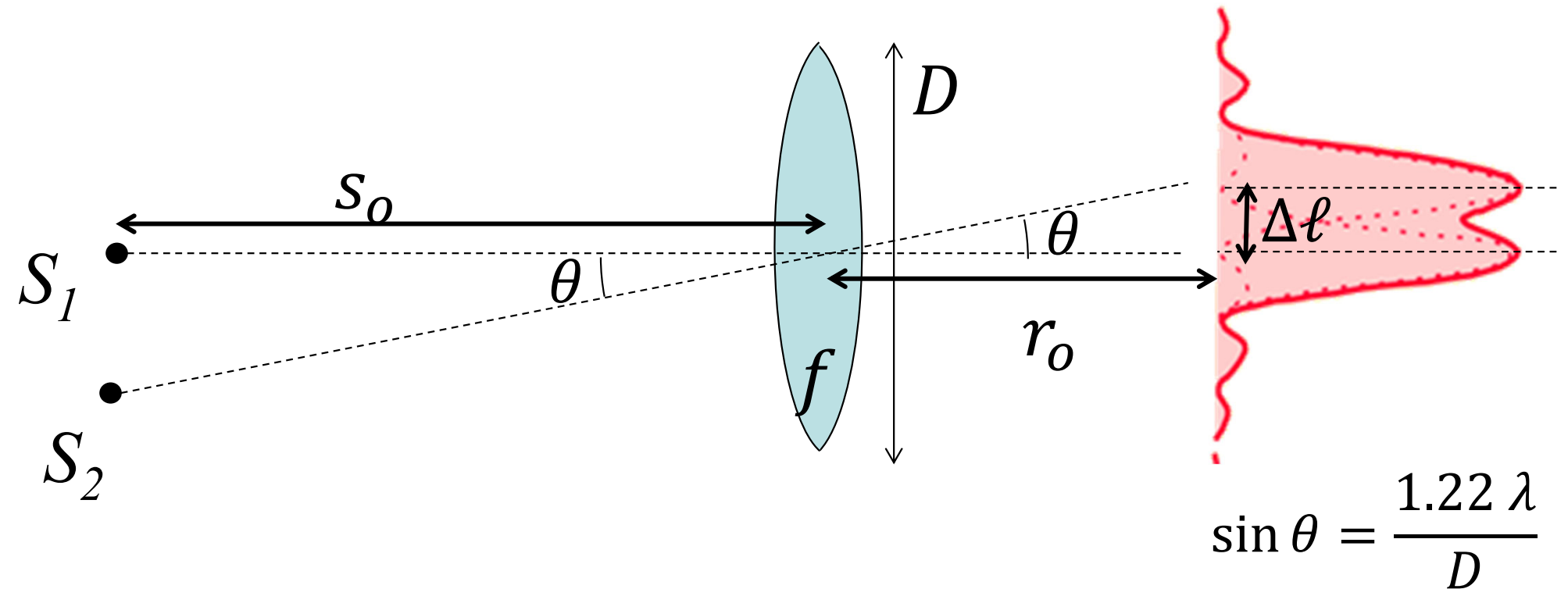
$$\sin \theta = \frac{1.22\lambda}{(2a)}$$

# Diffraction through a single circular aperture of diameter $D$ : limits on optical resolution



*When looking through an aperture, two objects are barely resolvable (distinguishable) when the center of one is at the first min of the other's diffraction pattern .*

# Resolution of imaging systems



*If sources are far away,  $r_0 \approx f$   
and  $\frac{\Delta \ell}{f} = \frac{1.22 \lambda}{D}$*